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DIVISION OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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ODD SEMESTER 2026-2027

END SEMESTER VIVA VOCE EXAMNIATION

HANDWRITING FORGERY DETECTION USING DEEP LEARNING: PAGE-LEVEL, MULTI-REFERENCE VERIFICATION

SDG Addressed: SDG 16 – Peace, Justice and Strong Institutions

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AGENDA

- Abstract
- Problem Definition, Scope And Objectives
- Literature Survey
- Proposed System Architecture
- Implementation
- Results And Discussion
- Conclusion
- References
- Outcome Specific Supporting Evidence

ABSTRACT

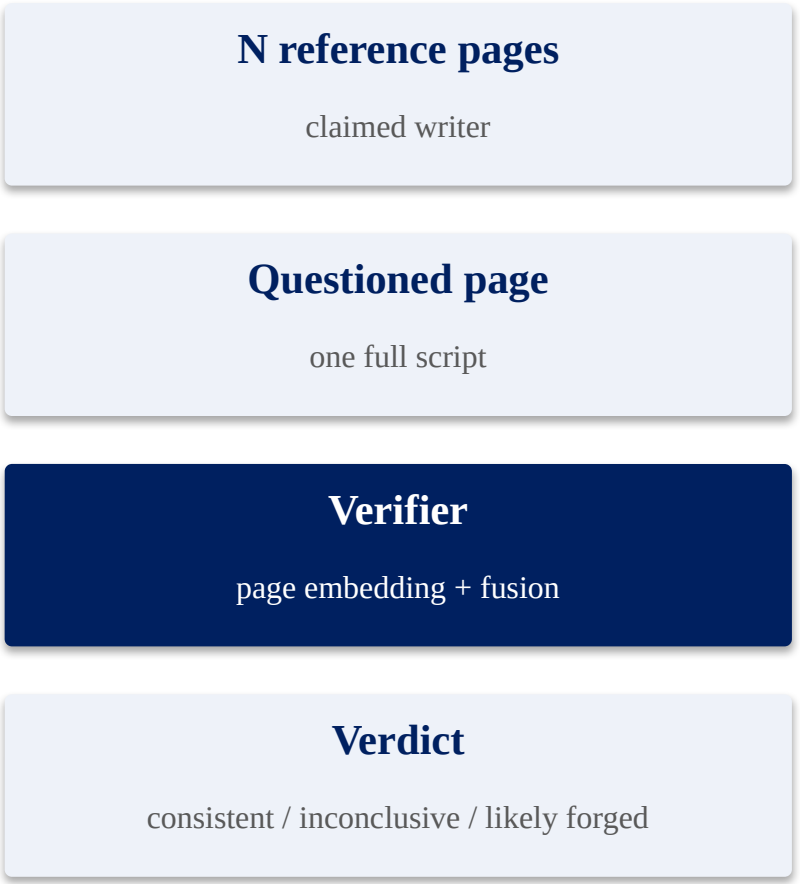
Deciding whether a questioned handwritten page matches a writer's earlier pages is a recurring problem in disputed exams, wills and contracts. We built a page-level verifier: a ResNet-18 trained from scratch embeds a whole page as an attention-pooled bag of ink patches, and one or several reference pages are fused into a calibrated verdict (consistent, inconclusive, likely forged).

Tested on held-out writers and on a corpus never used for tuning, it roughly halves the error of frozen DINOv2 features, but it is fooled by deliberate style change. No public data show one person imitating another, so these are not forgery detection rates.

2.35%	7.30%	3.3%	32%
IAM test EER	CVL test EER	Firemaker, natural writing	Firemaker, disguised

PROBLEM DEFINITION

- A questioned answer sheet: the owner offers earlier genuine scripts as references.
- The unit of evidence is a complete script, not a word, sentence or signature.
- Task: given N reference pages of a claimed writer and one questioned page, return a similarity score and a verdict.
- Hard because a writer varies (pen, mood, text), forgers imitate, and no public data show one person copying another.



SCOPE AND OBJECTIVES

In scope

- English full pages (IAM, CVL, Firemaker)
- 1 to 5 reference pages per writer
- Writer-disjoint test, cross-corpus test
- Calibrated three-way verdict, CPU demo

Out of scope (first paper)

- Four Indian languages (data collection paused)
- Real impersonation data (none public)
- Learned imitation generators
- Word-level and signature verification

Research questions

RQ1 How well does a page-level model separate writers, and does it beat untrained and frozen-feature baselines?

RQ2 Do extra reference pages help, and which fusion rule is best?

RQ3 Does it transfer to a new scanner and protocol (Firemaker), including disguised writing?

RQ4 Which training ingredients matter: distillation, pooling, content matching, mining, patches, augmentation?

LITERATURE SURVEY

Work	Level	What it does	Gap for our task
DeepWriter (Xing, Qiao 2016)	text patches	CNN for text-independent writer identification	identifies, does not verify a claim
FragNet (He, Schomaker 2020)	fragments	fragment network for writer identification	closed-set identification
Self-supervised ViT (Raven et al. 2024)	pages	unsupervised writer retrieval	ranks pages, no verdict
Attention verifier (Shaikh et al. 2020)	pairs	writer-independent verification	one pair, not many references
Self-supervised verification (Chauhan et al. 2024)	pairs	pre-training for verification	pairs, no calibration
SigNet (Dey et al. 2017)	signatures	Siamese signature verification	signatures, not full pages

Writer identification answers "which writer"; verification work compares pairs of words, patches or signatures. Little of it combines a whole questioned page with several references, reports calibrated support, or is allowed to say "inconclusive".

RESEARCH GAP AND OUR POSITION

Page-level

A complete script is the unit of evidence, embedded as an attention-pooled bag of 64 ink patches.

Multi-reference

One to five reference pages are fused (mean, max, top-2, centroid) and compared on one fixed set of writers.

Calibrated verdict

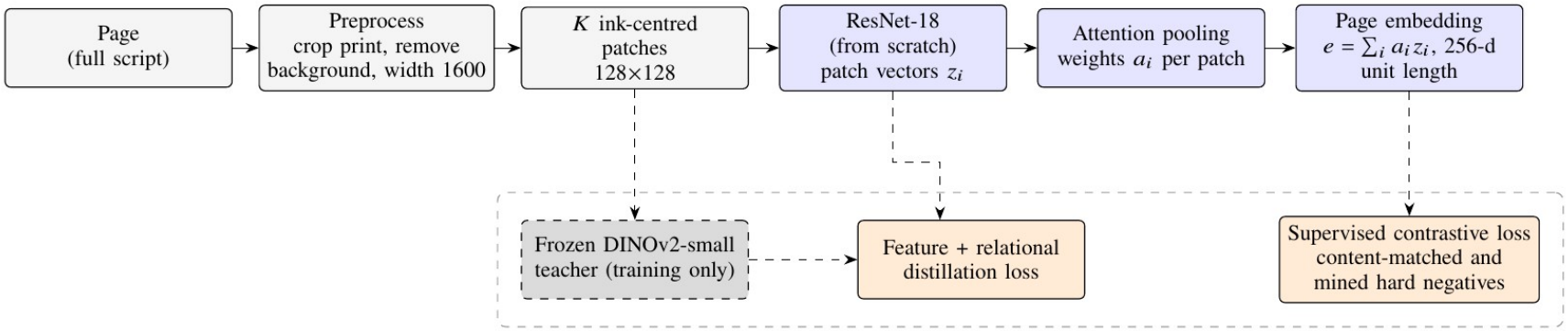
Thresholds fixed on validation data; the system may answer inconclusive.

To our knowledge no prior writer-verification work combines these elements; we do not claim any single component is new. Distillation and likelihood-ratio reporting each have precedent.

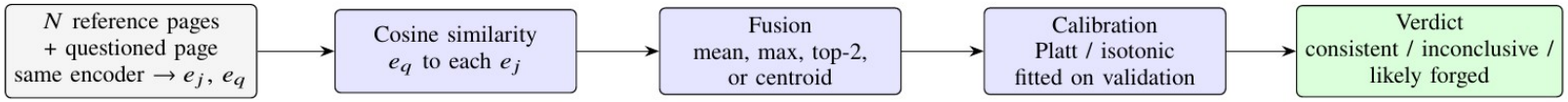
Honest protocol: writer-disjoint splits, test sets used once, a cross-corpus test never used for tuning.

PROPOSED SYSTEM ARCHITECTURE

(a) Page encoder and training



(b) Verification at inference (teacher and losses are not used)



(a) Training: each page becomes ink-centred 128 × 128 patches, encoded by a ResNet-18 trained from scratch and pooled by attention into a 256-d page embedding. A frozen DINOv2-small teacher is used only during training.

(b) Inference: the questioned page is compared with N references with the same encoder, scores are fused, calibrated and turned into a verdict.

MODEL AND TRAINING OBJECTIVE

Encoder	ResNet-18 (1 channel, random init) → 256-d patch vectors; attention pooling $a_i = \mathrm{softmax}(w^T \tanh(V z_i))$; page embedding is L2-normalised. 16 patches per page in training, 64 in evaluation.
Loss	Supervised contrastive loss ($\tau = 0.1$) on batches of 12 writers × 2 pages. 40% of batches are content-matched CVL batches (all writers copy the same texts); half use nearest-neighbour writers. Plus feature and relational distillation from frozen DINOv2-small (weights 0.5 + 0.5).
Verdict	Fuse N reference scores (mean, max, top-2, centroid), calibrate with Platt or isotonic regression fitted on validation only. Consistent if the score is at or above the 99th percentile of validation impostors; likely forged at or below the 1st percentile of genuine scores; otherwise inconclusive.

DATASETS AND PREPROCESSING

Corpus	Train writers / pages	Validation	Test
IAM	507 / 1262	50 / 190	99 / 392
CVL (English)	217 / 909	31 / 127	62 / 254
Firemaker	–	–	250 / 1000

- **IAM:** forms by many writers; the copy used is a third-party Kaggle one (official registration pending).
- **CVL:** all writers copy the same texts, which gives content-matched hard negatives. English texts only.
- **Firemaker:** test only, never used for tuning. Page 3 is self-disguise, not imitation of another person.
- **Preprocessing:** crop printed form parts, remove paper tone and scanner shading, resize to width 1600, drop blank and duplicate pages. Splits are writer-disjoint.

IMPLEMENTATION

Software

- Docker image handwriting-train:0.1 (PyTorch 2.6, CUDA 12.4)
- timm ResNet-18, DINOv2-small teacher, OpenCV, scikit-learn
- One module per stage: prep_cache, manifests, data, model, train, eval, multiref, verdict, report
- Unit tests (pytest) including a label-shuffle leak test
- FastAPI demo (CPU) and a live training dashboard

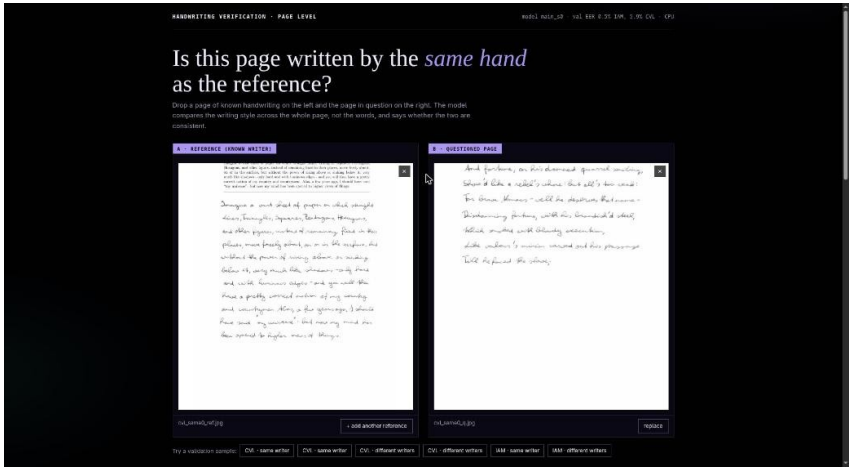
Training and hardware

- Laptop: RTX 4050 (6 GB), i7-13650HX, 15 GB RAM
- AdamW, lr 3e-4, cosine, up to 120 epochs, mixed precision, EMA 0.999
- Three seeds, about 50 min and 26 s per epoch each
- Seven 40-epoch ablations, one GPU job at a time
- Every table and figure is generated from the run files by report.py



Pipeline: each stage reads the previous stage's files, so any table can be regenerated.

LIVE DEMONSTRATION



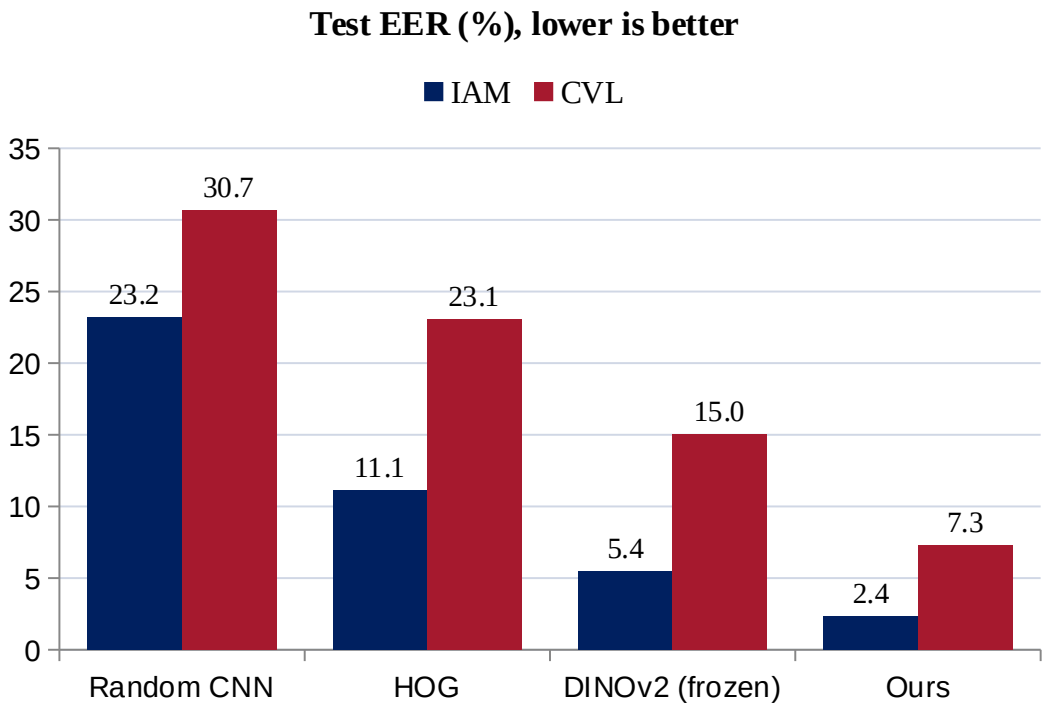
Upload a reference page and a questioned page



Verdict, score and where it falls among validation pairs

- Runs on CPU in about 2 to 3 seconds per comparison; one to five reference pages.
- Shows the calibrated verdict, similarity score, validation score distributions and attention heatmaps.
- Limits are printed on the page: it scores style similarity, not forgery directly.

RESULTS: MAIN COMPARISON



2.35 ± 0.12

IAM EER (%), AUC 0.996

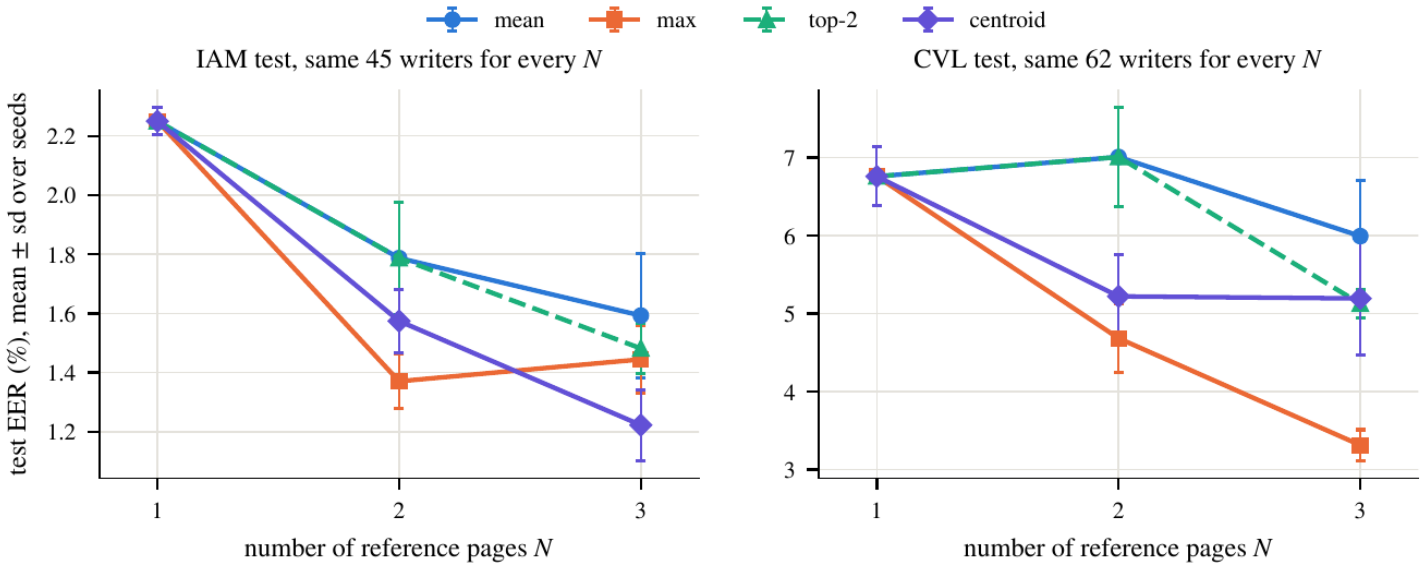
7.30 ± 0.29

CVL EER (%), AUC 0.968

Mean ± sd over three seeds, 99 IAM and 62 CVL test writers.

- The student halves the error of the frozen teacher on both corpora. Test is worse than validation (IAM 0.90 ± 0.31%), so validation numbers were optimistic.
- Strict operating point: at 1% false rejects, 80.2% of CVL impostors are accepted, so one page pair is weak evidence.

RESULTS: MORE REFERENCE PAGES

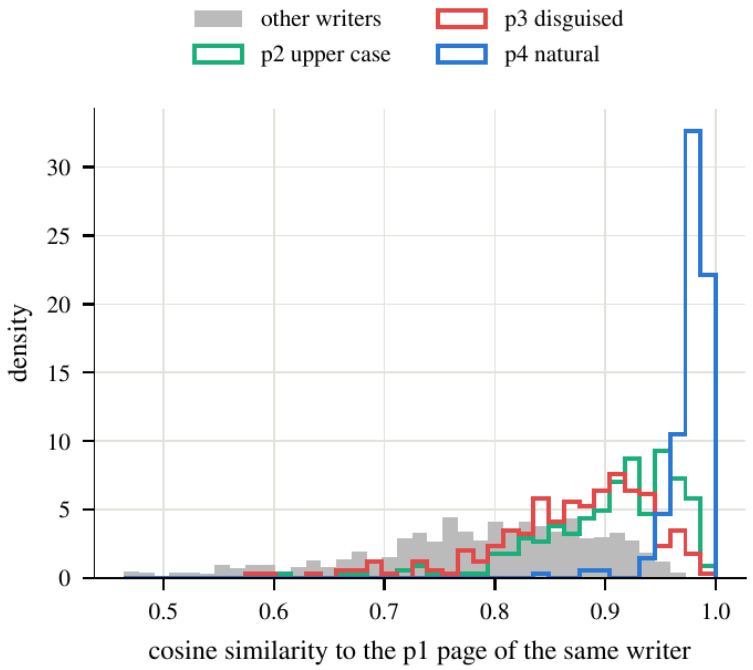


- Same writers for every N (45 IAM, 62 CVL). Centroid fusion: IAM 2.25 → 1.57 → 1.22; CVL 6.76 → 5.22 → 5.19 (N = 1, 2, 3).
- With five references, 17 IAM writers: 0.81 against 1.96 for one.
- On CVL the max rule is best at N = 3 (3.31), but the centroid was the rule chosen on validation, so this is a post-hoc observation.
- Bootstrap intervals overlap: the gain is consistent in direction but not established.

RESULTS: CROSS-CORPUS AND DISGUISE

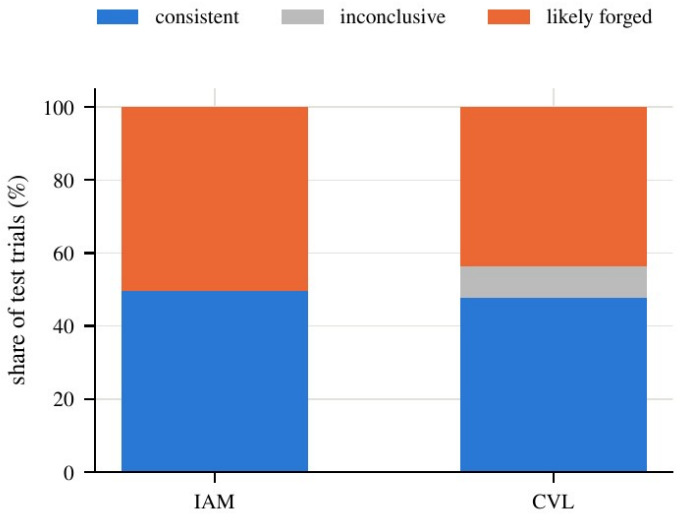
Firemaker pair	EER (%)	Same-writer cos.
p1 vs p4 (natural)	**3.33 ± 0.25**	0.978
p1 vs p2 (upper case)	**23.87 ± 0.47**	0.906
p1 vs p3 (disguised)	**32.00 ± 0.59**	0.874

- Natural writing transfers across corpora (3.3% EER) on 250 writers never used for tuning.
- Upper case and deliberate disguise break it: same-writer cosine falls from 0.98 to 0.87 while other writers stay near 0.79.
- A writer who changes style can be rejected, and a forger who copies a style is helped.



Seed 0, 250 writers

RESULTS: CALIBRATED VERDICTS



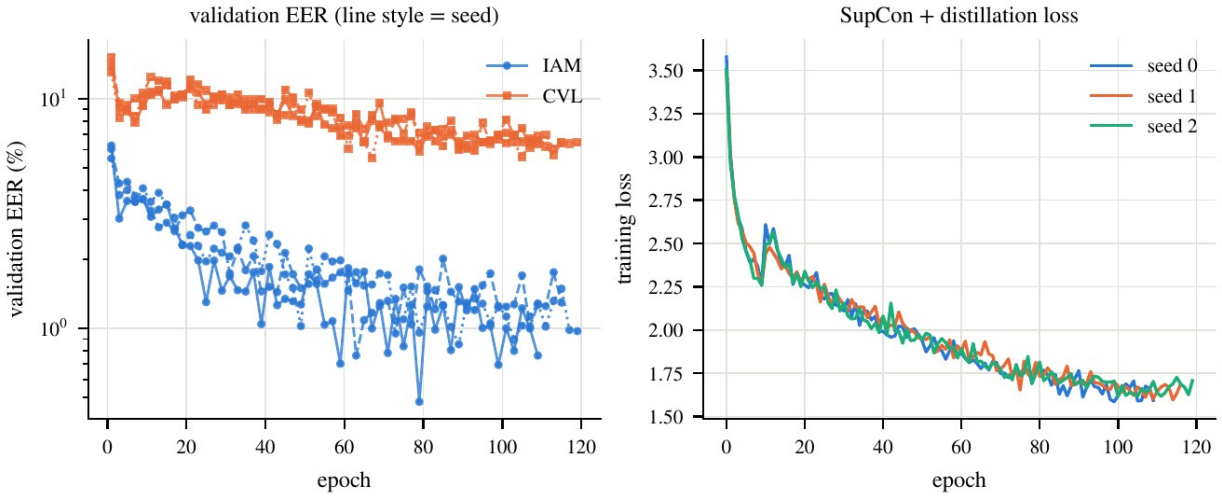
	CVL	IAM
Consistent (%)	47.8	49.7
Inconclusive (%)	8.5	0.0
Likely forged (%)	43.7	50.3
Error on decided (%)	2.8	0.7
ECE raw / Platt / isotonic	0.22 / 0.076 / 0.020	0.22 / 0.070 / 0.009

- Three references, thresholds from validation only. On CVL, 8.5% of trials are inconclusive and 2.8% of the decided ones are wrong.
- IAM is separable on validation, so one threshold is used and there is no inconclusive band; do not read this as evidence that the problem is easy.
- Raw cosine scores are not probabilities (ECE about 0.22); isotonic regression is the best calibrator.

ABLATIONS AND TRAINING BEHAVIOUR

Variant (40 epochs, 1 seed)	IAM	CVL	Mean
Full model (40 epochs)	1.72	8.98	5.35
No distillation	1.22	7.53	4.38
Mean pooling	2.47	8.10	5.29
No content-matched batches	1.71	7.42	4.57
No writer mining	2.05	8.45	5.25
8 patches per page	2.53	7.77	5.15
No augmentation	1.74	8.02	4.88

- Validation EER (%).
- No ingredient shows a clear benefit; three ablations beat the full 40-epoch model, but one seed cannot tell if that is real.
- Main 120-epoch runs reach 3.19 to 3.57 mean val EER.



LIMITATIONS

- **No real impersonation data.** Impostors are other writers, an easier case than a skilled forger.
- **Disguise breaks the model.** Firemaker p3 is self-disguise, reported as a stress test.
- **Synthetic pages are fonts.** Accepted for 0.03% of pairs on IAM thresholds; real unknown pages behave alike, so it says little about imitation.
- **Corpus effects.** IAM pages share scanner and pen, so IAM is optimistic; the IAM copy is third-party.
- **Scale.** English only, three seeds, one-seed ablations, one laptop GPU.



Attention maps are a visualisation only: dropping top-attended patches lowers the score by 0.0043 vs 0.0022 for random.

CONCLUSION AND FUTURE WORK

Conclusion

- A page-level verifier trained from scratch halves the error of frozen DINOv2 (2.35% IAM, 7.30% CVL EER).
- It transfers to a new corpus for ordinary writing (3.3%).
- Extra references and a calibrated, abstaining verdict help.
- It does not survive deliberate style change, and no ablated ingredient is shown to help.

Future work

- Volunteer-copied impersonation set per language
- Handwriting generators to test learned imitation
- Multi-seed ablations at the full schedule
- A larger validation set to choose the fusion rule
- Four Indian languages (data collection paused)

Page-level verification works for ordinary writing, but real impersonation data are needed before it can be called forgery detection.

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OUTCOME SPECIFIC SUPPORTING EVIDENCE



Page-Level Handwriting Forgery Detection:
Multi-Reference Verification with Calibrated Verdicts

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***Abstract*—Disputed handwritten scripts, such as an exam answer sheet suspected of being copied in another person's hand, require deciding whether a questioned page is consistent with a writer's earlier pages, yet most deep verifiers compare single words or signatures. We study page-level, multi-reference verification: a full page is embedded as an attention-pooled bag of ink patches by a ResNet-18 trained from scratch with content-matched hard negatives and distillation from a frozen DINOv2 teacher, and one or several reference pages are fused into a calibrated three-way verdict (consistent, inconclusive, likely forged). We train three seeds on IAM and CVL with writer-disjoint splits and test exactly once on held-out writers and on Firemaker, a separate corpus never used for training or tuning. The model reaches 2.35% EER on IAM and 7.30% on CVL, against 5.44% and 15.02% for frozen DINOv2 features, 11.1% and 23.1% for HOG, and 23.2% and 30.6% for an untrained network. On Firemaker, ordinary writing transfers across corpora (3.3% EER), but upper-case copying (23.9%) and deliberately disguised writing (32.0%) break it. On one fixed set of writers, three reference pages lower the centroid-fusion EER from 2.25% to 1.22% on IAM and from 6.76% to 5.19% on CVL, and the maximum rule alone reaches 3.31% on CVL, a post-hoc observation. A verdict calibrated on validation data leaves 8.5% of CVL trials inconclusive and is wrong on 2.8% of the rest. No ablated training ingredient shows a clear benefit in a short, single-seed, 40-epoch comparison. No public data show one person imitating another, so these are not forgery detection rates, and the font-rendered pages we tested say nothing about learned imitation.**

***Index Terms*—handwriting verification, forgery detection, writer verification, metric learning, knowledge distillation, calibration**

I. Introduction

Handwritten examination scripts, wills, contracts and notes are routinely disputed. A typical case is an answer sheet whose ownership is questioned because another person is suspected of copying the answers in an imitation of the owner's handwriting. The owner can then offer earlier genuine scripts as references, and an examiner must decide whether the questioned script is consistent with them. This is a *page-level, multi-reference* verification problem: the evidence is a complete script, not a single word, sentence or signature, and several known pages are available.

Most deep-learning work on handwriting is organised differently. Writer identification and retrieval systems assign a page to one of many known writers or rank a collection by similarity [1], [2], [3], [4], [5]. Verification work compares a pair of samples, usually words, text patches or signatures [6], [7], [8], [9]. Little of it combines a whole questioned page with several reference pages, reports a calibrated degree of support, or allows the system to answer "inconclusive".

A further difficulty is the threat model. Public datasets contain genuine pages by many writers, but almost none contain a person deliberately imitating another. At the same time, generative models can now imitate a given writing style [10], [11], [12], [13].

This paper builds and evaluates a page-level verifier under these conditions. Our contributions are:

- A page-level verifier that embeds a full script as an attention-pooled bag of ink patches and compares a questioned page with one or several reference pages, with four fusion rules evaluated.
- Content-matched hard negatives on CVL (different writers copying the same text), a calibrated three-way verdict with thresholds fixed on validation data, and a compact ResNet-18 student distilled from a frozen self-supervised ViT; an ablation shows no clear benefit of any single ingredient in a short schedule.
- An evaluation protocol with writer-disjoint splits, a cross-dataset test on a corpus never used for training or tuning, a controlled multi-reference study, and explicit reporting of what the available data cannot show, including that deliberate style change breaks the model.

To our knowledge, no prior writer-verification work combines these elements, but we do not claim that any single component is new. Distillation and likelihood-ratio reporting each have precedent [14], [15].

We study four research questions. **RQ1:** how well does a page-level model separate writers, and does it beat untrained and frozen-feature baselines? **RQ2:** does adding reference pages help, and which fusion rule is best? **RQ3:** does the model transfer to a corpus recorded with a different scanner and protocol (Firemaker), including pages written in a disguised style? **RQ4:** which training ingredients matter: distillation, attention pooling, content-matched batches, writer mining, patch count and augmentation?

II. Related Work

Writer identification and retrieval. Text-independent writer identification with CNNs was established by DeepWriter [1], and later work introduced fragment-based and attention-based

First page of the paper draft

Selected outcome: Journal publication

- Paper draft: 8 pages, IEEEtran, 270-word abstract, six result figures and eight tables; submission venue not yet chosen.
- Every number is generated from the run files and mapped to its evidence file.
- Code, unit tests, run logs and Docker setup are in the project repository.
- Working demo and live training dashboard.
- Pending: citation check, author block confirmation, official IAM registration.